

Formation Engine Canonical Simulation Methodology

Condensed Reference (12 pages)

Version 3.0.0 — January 2025

Covers §0–§13 of the full 186-page methodology

Contents

1 Foundational Thesis and Scope (§1)

1.1 The Problem

Modern computational materials science has a gap at intermediate scales (100–10,000 atoms). Quantum methods are too expensive; continuum models assume structure rather than generating it. The Formation Engine fills this gap: a reproducible instrument for generating physically admissible structures under explicit assumptions.

It is positioned as an *instrument*, not a prediction claim—analogous to a microscope that enables observation rather than asserting theory.

1.2 Four Axioms

1. **Explicit units everywhere.** Positions in Å, energies in kcal/mol, temperatures in K. No reduced units, no hidden normalization.
2. **No silent domain switching.** The force field, integrator, and boundary conditions are declared upfront and do not change mid-simulation.
3. **Deterministic core.** Given identical state, the trajectory is identical. All randomness is controlled by the seed.
4. **Periodic table as sole authority.** No molecular databases, no empirical crystal structures. Parameters derived from Z , atomic weight, covalent/VDW radii, electronegativity, ionization energy, and electron affinity via UFF.

1.3 Physical Admissibility

A configuration is *physically admissible* if it satisfies: (1) bounded energy evolution, (2) continuous forces below 10^3 kcal/(mol·Å), (3) consistent boundary behaviour, (4) charge conservation $|Q_{\text{total}}| < 10^{-6} e$, and (5) qualitative stability under timestep halving.

Admissibility does *not* imply thermodynamic equilibrium, synthetic feasibility, or experimental observability.

1.4 Explicit Scope Boundaries

Extension points (quantum corrections, reactive potentials, ML surrogates) are architecturally reserved but not implemented in v3.0.0.

2 Identity–State Decomposition (§0, §2)

2.1 Particle Identity Vector (§0)

Each particle carries a time-invariant identity:

$$\mathbf{I}_i = (Z_i, A_i, Q_i, \Sigma_i, \Lambda_i, \Theta_i) \in \mathbb{R}^6 \quad (1)$$

- $Z_i \in \mathbb{N}^+$: atomic number (immutable identity anchor)
- $A_i \in \mathbb{R}^+$: mass (isotopic or averaged)
- $Q_i \in \mathbb{R}$: effective charge participation coefficient
- $\Sigma_i \in \{0, 1, 2, 3, 4\}$: structural role (inert/ionic/covalent/metallic/mixed)
- $\Lambda_i \in \{0, 1, 2, 3\}$: stability class (transient/metastable/stable/bulk)
- $\Theta_i \in \mathbb{B}^n$: provenance hash (transformation lineage)

The Canonical Particle Container Postulate extends this to $\mathcal{C}_i = (\mathbf{I}_i, \mathbf{G}_i, \mathbf{C}_i^{\text{comp}})$, where $\mathbf{G}_i$ is a quantum descriptor bundle (spin, colour, flavour—reserved for future use in the classical framework) and $\mathbf{C}_i^{\text{comp}}$ carries computational annotations (complexity class, kernel selector, solver priority).

2.2 The State Tuple (§2)

$$\mathcal{S}(t) = (N, \mathbf{M}, \boldsymbol{\tau}, \mathbf{X}, \mathbf{V}, \mathbf{Q}, \mathbf{B}, \mathbf{F}, \mathcal{E}, \mathcal{L}, \mathcal{B}) \quad (2)$$

Tripartite classification:

Category	Fields	Mutability
Identity	$N, \mathbf{M}, \boldsymbol{\tau}$	Immutable after construction
Phase	$\mathbf{X}, \mathbf{V}, \mathbf{Q}, \mathbf{B}$	Evolve under equations of motion
Scratch	$\mathbf{F}, \mathcal{E}, \mathcal{L}, \mathcal{B}$	Recomputed from phase; never persisted independently

This decomposition enforces consistency: forces are zeroed before every evaluation, masses are locked to the periodic table via $\boldsymbol{\tau}$, and topology is inferred from geometry rather than stored as input.

2.3 Energy Ledger

The potential energy decomposes into physically distinct terms:

$$\mathcal{E} = (U_{\text{bond}}, U_{\text{angle}}, U_{\text{torsion}}, U_{\text{vdW}}, U_{\text{Coulomb}}, U_{\text{external}}) \quad (3)$$

Every force model reports its contribution to each term separately. This reveals *why* a structure formed—whether it is bond-dominated ($U_{\text{bond}} \gg U_{\text{vdW}}$) or dispersion-dominated ($U_{\text{vdW}} \gg U_{\text{bond}}$).

2.4 File Format Hierarchy

Format	Extension	Content
XYZ	.xyz	Static geometry: N + comment + element/position lines
XYZA	.xyza	Animated trajectory: sequential XYZ frames with timestamps
XYZC	.xyzc	Checkpoint: positions + velocities + $(T, P, E_{\text{pot}}, E_{\text{kin}})$ + SHA-256 provenance hash

Each level adds information; each is self-contained. The XYZC format enables exact trajectory continuation.

3 Physical Interaction Model (§3)

3.1 Interface

All force models implement a pure function:

$$\text{eval} : \mathcal{S} \times \Theta \rightarrow (\mathbf{F}, \mathcal{E}) \quad (4)$$

No hidden state. No history dependence. Same inputs always yield same outputs.

3.2 Nonbonded Potentials

Lennard-Jones 12-6:

$$U_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (5)$$

Combining rules (Lorentz–Berthelot): $\sigma_{ij} = (\sigma_i + \sigma_j)/2$, $\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}$. Parameters from UFF: $\sigma = 2^{5/6} r_{\text{vdW}}$, ε from UFF well depths.

Quintic switching function between $r_{\text{on}} = 9 \text{ \AA}$ and $r_{\text{cut}} = 10 \text{ \AA}$ ensures C^2 continuity at the cutoff.

Coulomb electrostatics:

$$U_{\text{Coul}}(r) = \frac{k_e q_i q_j}{r}, \quad k_e = 332.0636 \text{ kcal} \cdot \text{\AA} / (\text{mol} \cdot e^2) \quad (6)$$

Same switching function applied. **Known limitation:** Coulomb–integrator coupling produces energy drift in MD for ionic systems; use FIRE minimization only for charged systems in v3.0.0.

3.3 Bonded Potentials

$$U_{\text{bond}} = k_b (r - r_0)^2 \quad k_b \in [200, 600] \text{ kcal}/(\text{mol} \cdot \text{\AA}^2) \quad (7)$$

$$U_{\text{angle}} = k_\theta (\theta - \theta_0)^2 \quad k_\theta \in [40, 120] \text{ kcal}/(\text{mol} \cdot \text{rad}^2) \quad (8)$$

$$U_{\text{torsion}} = V_n [1 + \cos(n\phi - \gamma)] \quad V_n \in [0.5, 5] \text{ kcal/mol} \quad (9)$$

$$U_{\text{improper}} = k_{\text{imp}} (\psi - \psi_0)^2 \quad (10)$$

Bond inference: atoms i, j are bonded if $r_{ij} < f \cdot (R_i^{\text{cov}} + R_j^{\text{cov}})$ with tolerance $f = 1.2$.

4 Thermodynamic Framework (§4)

4.1 Unit System

Quantity	Unit	SI Relation
Distance	$\text{\AA}$	10^{-10} m
Mass	amu	$1.66054 \times 10^{-27} \text{ kg}$
Time	fs	10^{-15} s
Energy	kcal/mol	$4184/N_A \text{ J per particle}$
Temperature	K	—
Charge	e	$1.602 \times 10^{-19} \text{ C}$

4.2 Conversion Constants (As Implemented)

$$k_B = 0.0019872041 \text{ kcal}/(\text{mol} \cdot \text{K}) \quad (11)$$

$$C_{\text{KE}} = 2390.057361 \text{ (amu} \cdot \text{\AA}^2/\text{fs}^2 \rightarrow \text{kcal/mol}) \quad (12)$$

$$\text{ACC_CONV} = C_{\text{KE}}^{-1} \text{ (kcal}/(\text{mol} \cdot \text{\AA}) \rightarrow \text{amu} \cdot \text{\AA}/\text{fs}^2) \quad (13)$$

4.3 Observables

Temperature from equipartition:

$$T = \frac{2K}{N_{\text{df}} k_B}, \quad N_{\text{df}} = 3N - 3 \quad (14)$$

Kinetic energy:

$$K = C_{\text{KE}} \sum_{i=1}^N \frac{1}{2} m_i |\mathbf{v}_i|^2 \quad (15)$$

Maxwell-Boltzmann initialization: $v_{i,\alpha} \sim \mathcal{N}(0, \sqrt{k_B T/m_i})$ followed by centre-of-mass velocity subtraction.

5 Time Integration (§5)

5.1 Velocity Verlet (NVE)

The default integrator. Four-step algorithm at each timestep Δt :

$$\mathbf{v}_i(t + \frac{\Delta t}{2}) = \mathbf{v}_i(t) + \frac{\mathbf{F}_i(t)}{m_i} \text{ACC_CONV} \frac{\Delta t}{2} \quad (16)$$

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{v}_i(t + \frac{\Delta t}{2}) \Delta t \quad (17)$$

$$\mathbf{F}_i(t + \Delta t) = -\nabla_{\mathbf{x}_i} U(\mathbf{X}(t + \Delta t)) \quad (18)$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t + \frac{\Delta t}{2}) + \frac{\mathbf{F}_i(t + \Delta t)}{m_i} \text{ACC_CONV} \frac{\Delta t}{2} \quad (19)$$

Symplectic (preserves phase-space volume), time-reversible, 2nd-order. Default $\Delta t = 1$ fs.

Validation: Energy drift $< 3 \times 10^{-5}$ kcal/mol/atom over 10^4 steps ($33\times$ safety margin). Reversibility RMSD = 2.3×10^{-9} Å.

5.2 Langevin Dynamics (NVT)

Euler-Maruyama discretisation of the Langevin equation:

$$m_i \dot{\mathbf{v}}_i = \mathbf{F}_i - \gamma m_i \mathbf{v}_i + \sqrt{2\gamma m_i k_B T} \boldsymbol{\xi}_i(t) \quad (20)$$

The friction coefficient γ (ps^{-1}) couples the system to an implicit heat bath. Random forces $\boldsymbol{\xi}_i$ are drawn from seeded `std::mt19937`.

Validation: Temperature error $< 0.6\%$ at $T = 300$ K for Ar_{100} .

5.3 FIRE Minimisation

Fast Inertial Relaxation Engine: modified MD where velocity is projected onto the force direction and Δt adapts.

1. Compute $P = \mathbf{F} \cdot \mathbf{v}$
2. If $P > 0$: mix velocity toward force direction, increase Δt
3. If $P \leq 0$: zero velocity, decrease Δt , reset mixing

Convergence: $\max |\mathbf{F}_i| < \varepsilon_F$ (default $0.01 \text{ kcal}/(\text{mol} \cdot \text{\AA})$).

6 Formation Physics (§6)

A **formation** is a local minimum of the potential energy surface reached through a physically motivated search:

1. **Initialise:** VSEPR geometry prediction from valence electron count and lone-pair repulsion model
 2. **Explore:** Short MD runs at elevated temperature cross energy barriers
 3. **Quench:** FIRE minimisation projects away thermal noise, revealing local minima
 4. **Score:** Multiplicative 6-factor model ranks formations
- VSEPR energy:**

$$U_{\text{VSEPR}} = k_{\text{VSEPR}} \sum_{a < b} \frac{w_{ab}}{[\varepsilon + (1 - \cos \theta_{ab})]^p} \quad (21)$$

Domain weights: LP-LP = 2.0, LP-BP = 1.5, BP-BP = 1.0 (encodes empirical repulsion ordering).

Bond inference: Mutable cache, re-inferred on demand via covalent radii, never serialised as truth. Stored as `inferred_bonds` with `bonds_computed` flag.

7 Statistical Interpretation (§7)

7.1 Online Statistics

Welford’s one-pass algorithm avoids catastrophic cancellation in variance computation:

$$\bar{x}_n = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n} \quad (22)$$

$$M_{2,n} = M_{2,n-1} + (x_n - \bar{x}_{n-1})(x_n - \bar{x}_n) \quad (23)$$

Variance: $\text{Var}(x) = M_2/(n-1)$.

7.2 Stationarity Gate

Convergence criterion: coefficient of variation $\text{CV} = \sigma/|\mu|$ drops below threshold ϵ for 10 consecutive windows. Prevents premature termination from transient plateaux.

7.3 Kabsch Alignment

Optimal rigid-body superposition via SVD:

$$\mathbf{R}^* = \arg \min_{\mathbf{R} \in \text{SO}(3)} \sum_{i=1}^N |\mathbf{R} \mathbf{x}_i^{\text{ref}} - \mathbf{x}_i|^2 \quad (24)$$

Compute $\mathbf{H} = \mathbf{X}^{\text{ref},T} \mathbf{X}$, SVD $\mathbf{H} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$, then $\mathbf{R} = \mathbf{V} \text{diag}(1, 1, \det(\mathbf{V} \mathbf{U}^T)) \mathbf{U}^T$.
RMSD after alignment quantifies structural similarity.

8 Reaction & Electronic Properties (§8–9)

8.1 QEq Charge Equilibration (§9)

Minimise the energy functional:

$$E(\{q_i\}) = \sum_{i=1}^N (\chi_i q_i + \eta_i q_i^2) + \sum_{i < j} \frac{k_e q_i q_j}{r_{ij}} \quad (25)$$

subject to $\sum_i q_i = Q_{\text{total}}$, where $\chi_i = (\text{IE}_1 + \text{EA})/2$ (Mulliken electronegativity) and $\eta_i = (\text{IE}_1 - \text{EA})/2$ (chemical hardness).

This yields a linear system $(2\boldsymbol{\eta} + \mathbf{J})\mathbf{q} = \boldsymbol{\chi} + \lambda \mathbf{1}$ solved for equilibrium charges.

8.2 Reactivity Indices (§9)

Fukui functions (finite-difference approximation):

$$f_i^+ = q_i(N) - q_i(N+1) \quad (\text{electrophilic attack}) \quad (26)$$

$$f_i^- = q_i(N-1) - q_i(N) \quad (\text{nucleophilic attack}) \quad (27)$$

8.3 Reaction Prediction (§8)

HSAB (Hard-Soft Acid-Base) matching identifies compatible reactive sites. Activation barriers estimated via Bell–Evans–Polanyi:

$$E_a = E_a^0 + \alpha \Delta H \quad (28)$$

Reaction templates (S_N2 , addition, elimination) encode topology changes. Status: interface defined, full MD integration pending.

9 Multiscale (§10)

Scale	N	Method
Molecular	2–50	Full MM + FIRE
Cluster	50–500	MD + quench
Bulk	500–10k	PBC + supercell replication
Coarse-Grained	10k+	CG placeholder

Supercell construction: $\mathbf{r}_{i,pqr} = \mathbf{r}_i + p\mathbf{a} + q\mathbf{b} + r\mathbf{c}$ for $p \in [0, n_a)$, etc. Bond graph re-inferred from supercell geometry (not replicated from unit cell) as a validation sanity check.

Provenance chain: `unit_cell.xyz` $\rightarrow$ `supercell(2,2,2)` $\rightarrow$ `relax(FIRE)` $\rightarrow$ `final.xyzC`.

10 Self-Audit Infrastructure (§11)

10.1 Determinism Contract

Given (formula, seed, parameters), the engine must produce **bit-identical output**. Enforced by:

- Seeded RNG (`std::mt19937`)
- Index-ordered force evaluation (no thread-dependent accumulation)
- IEEE 754 compliance (no `-ffast-math`)
- No mutable global state

10.2 Three Audit Tools

Failure Classifier (`tools/failure_classifier.py`, 250 lines):

- 6 categories: NUMERICAL, PHYSICS, OUT-OF-DOMAIN, CONVERGENCE, TIMEOUT, UNKNOWN
- UNKNOWN budget < 5%
- Each failure tagged with minimal reproduction command

Gap Targeter (`tools/gap_targeter.py`, 200 lines):

- $6 \times 10 \times 10 = 600$ cells over (element group, temperature, density)
- Identifies sparse coverage and high-mismatch regions
- Target: 95% cell coverage

Regression Detector (`tools/regression_detector.py`, 200 lines):

- 4 invariants: energy monotonicity, bond stability, score boundedness, classification consistency
- Binary verdict: pass / review required

11 Validation Doctrine (§12)

35 hierarchical tests across 5 levels. Each test has an explicit pass/fail criterion defined *before* execution.

Level	Category	Tests	Pass	Key Criterion
0	Unit system	12	12/12	Constants match literature to 10^{-6}
1	Force evaluation	8	8/8	Analytic vs. numerical gradient $< 10^{-5}$
2	Integration	5	5/5	Energy drift $< 10^{-3}$ kcal/mol/atom
3	Thermodynamics	8	7/8	$ \langle T \rangle - T_{\text{target}} /T_{\text{target}} < 5\%$
4	Reproducibility	3	3/3	Bit-identical across seeds
Total		36	35/36	97%

Known failure: Level 3 Langevin temperature target for small systems (finite-size thermostat coupling).

Production certified for: Noble gases (Ar, Xe, Kr), hydrocarbons (CH_4 , benzene, alkanes), small organics (H_2O , NH_3 , CH_3OH), molecular clusters.

Requires fix before production: Ionic MD (NaCl , MgO) — use FIRE only until Coulomb coupling resolved.

12 Future Work (§13)

Ordered by impact-to-effort ratio:

1. **Neighbour lists** — Cell-linked or Verlet lists for $O(N)$ force evaluation ($50\times$ speedup for $N > 1000$)
2. **NPT barostat** — Pressure control via Parrinello–Rahman for crystal structure relaxation
3. **Reactive bond orders** — ReaxFF-style continuous bond order for dynamic bond breaking/formation
4. **Quantum corrections** — Delta-learning: $U = U_{\text{classical}} + \Delta U_{\text{QM}}$ from single-point DFT
5. **ML potentials** — SchNet/MACE surrogates trained on formation engine ensembles
6. **Grand canonical** — μVT ensemble for variable-composition systems
7. **Explicit solvent** — TIP3P/SPC/E water models

All extensions preserve the core contract: determinism, explicit units, provenance tracking, validation tests.

Full methodology: 11 L^AT_EX source files in `docs/`.

Repository: <https://github.com/LMSM3/VSEPR-SIM>

License: MIT | 821 source files | 35 validation tests